

🔔 Don't forget:

Academia Premium 30 day trial for just \$1

Upgrade now for \$1

×

A

🔍 Search

▼

99

Mentions

📊

Analytics

⬆

Upload

🛠

Tools

Try Premium for \$1

99+

 Nathan

▼

Heraclitean Extended Relational Field Dynamics: Filament Ontology, Emergent Curvature, Bridge Maps to Effective Geometry and Quantum Sectors, and the Emergent Origin of the Planck Scales

By Stephen Andreu

👁 3 Views

📄 42 Pages

🔗 1 File

▼

🔗 [Physics](#), [Theoretical Physics](#), [Quantum Physics](#), [Dynamical Systems](#), [Quantum Gravity](#), [Show more](#)

This paper presents Heraclitean Extended Relational Field Dynamics (HERFD), a speculative but mathematically structured research program in which physical spacetime is not fundamental and the primitive entities are interacting filaments carrying internal fields, director frames, and topological linking structure. The framework rests on five conceptual postulates: no physical spac ... [Read more](#)

⬇ Download PDF

🔒

Full PDF Package

🔒

Translate

⋮

Original PDF	Related
--------------	---------

to collective coordinates and low-energy bosonic fluctuation sectors.

The relational correlation length ℓ_{corr} is defined as the inverse square-root of the spectral gap of the relational graph Laplacian, a genuine gauge-invariant HERFD-internal quantity that is dimensionless before unit-fixing and acquires the dimension of length only through three emergent quantities in the Planck-matched regime: the collective action quantum $\hbar_{\text{eff}}^-$ (which sets the action scale), the emergent cone speed c_{eff} (which sets the causal conversion scale), and the induced gravitational coupling G_{eff} or equivalently U_0 (which sets the gravitational scale). The ultraviolet

Andreu — HERFD

April 2026

cutoff scale ℓ_{UV} is introduced as a conceptually distinct quantity: the continuum description breaks down at ℓ_{UV} , and the identification $\ell_{UV} \sim \ell_{\text{corr}}$ holds only in the Planck-matched regime, stated explicitly as a matching condition rather than a definitional equivalence. All four dimensionless scale parameters $\alpha_a, \alpha_R, \alpha_H, \alpha_{ij}$ are classified precisely. The continuum-limit tensors K_{ij} and I_c , defined in eq. (56), are held fixed in the continuum limit; α_R is fixed by a renormalization-group optimization condition; α_H is bounded by chart admissibility; and α_{ij} is physically irrelevant in the smooth regime. The integers m_{ij}^* labeling the linking potential are reclassified as topological superselection sector labels rather than free model parameters.

parameters.

A concrete single-cone strictly hyperbolic low-frequency regime is proved for the bosonic tetrahedral-lattice sector, and the emergent cone speed $c_{\text{eff}} = \sqrt{K_{ij}\xi^i\xi^j}/I_c$ is derived as a HERFD-internal quantity that is independent of the lattice regulator in the physical continuum limit. The conformal normalization $\Omega^2(F) = U(F)/U_0$ is derived as the unique factor consistent with background flatness and Einstein-frame reduction at background order, where U_0 is obtained by a one-loop Seeley-DeWitt heat-kernel calculation. Crucially, the one-loop result is presented as a *self-consistency relation* relating U_0 , the UV identification ℓ_{UV} , and the Planck-matched regime, rather than an independent first-principles derivation; the species consistency condition $N_\theta \approx 12\pi/f_{\text{tot}}(\alpha_R^*, \beta_{\text{UV}})$ is accordingly classified as a *leading-order species consistency relation* implied by the current one-loop truncation, where $f_{\text{tot}} = f(\alpha_R^*)/\beta_{\text{UV}}^2$ absorbs both the regularization scheme factor and the UV-matching ambiguity. The effective Planck length $\ell_{P,\text{eff}} = \sqrt{\bar{h}_{\text{eff}}G_{\text{eff}}/c_{\text{eff}}^3}$ is defined by three emergent quantities: $\bar{h}_{\text{eff}}$, c_{eff} , and G_{eff} (equivalently U_0). The effective gravitational field equation derived from the coarse-grained action yields, in the background limit, an Einstein equation with an effective cosmological term, not the sourceless Einstein equation. Wavefunction renormalization and effective mass are expressed in terms of $U(F)$, determined at background order. On-shell covariant conservation is established from the variational principle.

The result is not a finished theory of gravity or microscopic quantum matter. It is a research program intended to make the relational ontology operational, remove hidden-background ambiguities, derive the microscopic dynamics explicitly, and show how effective geometry and controlled quantum sectors arise from deeper filament dynamics.

Contents

1	Introduction	5
2	Conceptual Postulates of HERFD	7

2	Conceptual Postulates of HERFD	4
2.1	Postulate I: Filamentary Relational Ontology	7
2.2	Postulate II: Auxiliary Similarity Arena and Gauge Reduction	8
2.3	Postulate III: Heraclitean Generic Non-Repeatability	8
2.4	Postulate IV: Time Emerges from Change	8
2.5	Postulate V: Bridge to Effective Geometry and Quantum Sectors	8
2.6	Internal Definition of the Relational Correlation Length and the Unit-Fixing Protocol	9
2.7	Classification of the Dimensionless Scale Parameters.	10
3	Microscopic Configuration Space	11
3.1	Representative Filament Data	11
3.2	Redundancy Group and Reduced State Space	11
3.3	Scale-Covariant Local Reference Lengths and Intrinsic Relational Observables	12
3.4	Framed Filament Strain Variables	12
4	Heraclitean Fingerprints and Local Spectral Data	13
4.1	Kernel Moments	13
4.2	Local Relational Laplacian	13
4.3	Fingerprint Field and Separation Criterion	13
5	Core Action and Explicit Microscopic Dynamics	14
5.1	Supermetric on Reduced Configuration Space	14
5.2	Potential Sector and Topological Superselection	15
5.3	Jacobi-Type Action and Emergent Time	16
5.4	Constraint and Equations of Motion	16
6	Emergent Causal Structure, Curvature, and Effective Gravitational Action	17
6.1	Linearized Perturbations and the Genuine Principal Operator	17
6.2	Hyperbolicity, the Emergent Conformal Metric Class, and Its Normalization at Background Order	17
6.3	Coarse-Grained Charts, Frames, and Connection Data	19
6.4	Heraclitean Stress and the Bosonic Matter Sector	19
6.5	Effective Coarse-Grained Action	20
6.6	Metric Variation and the Effective Gravitational Field Equation	20
6.7	Fingerprint Variation and On-Shell Conservation	20

7	Explicit 2+1 Lattice Model	21
7.1	Microscopic Triangular Lattice History	21
7.2	Coarse-Grained Coefficients	21
8	Explicit Bridge Map for the 2+1 Lattice Model	21
8.1	Temporal Leg	21
8.2	Heraclitean Chart Leg	22
8.3	Metric, Frame, and Bosonic Matter Legs	22
8.4	Composite Bridge Map	22
9	The 3+1 Lattice Model	22
9.1	Tetrahedral Bond Set and Microscopic History	22
9.2	Discrete Action for the Bosonic Tetrahedral Sector	22
9.3	Continuum Coefficients	23
10	Explicit Bridge Map for the 3+1 Sector	23
10.1	Uniform Positivity Hypotheses and the Correct Continuum Limit Scaling	23
10.2	Concrete Hyperbolicity Theorem and Emergent Speed of Light	24
10.3	Continuum-Limit Theorem	26
10.4	Chart Leg, Frame Leg, and Composite Classical Bridge	27
11	Bridge to Effective Quantum Theory of Collective Modes	27
11.1	Quadratic Expansion about a Metastable History	27
11.2	Canonical Momentum and Symplectic Form	28
11.3	Finite-Dimensional Collective-Coordinate Bridge	28
11.4	Field-Theoretic Quantum Bridge	28
11.5	Conditional Spinorial Sector	29
11.6	Honest Claim Boundary	29
11.7	Emergent Quantum of Action and Gauge-Invariance	29
11.8	One-Loop Self-Consistency Derivation of the Induced-Gravity Coefficient U_0	30
12	Why Filaments	33
13	Limitations and Methodological Cautions	34
14	Conclusion	36
A	Compact List of Principal Formulas	37
	References	39

*Andreu — HERFD**April 2026*

1. Introduction

The aim of Heraclitean Extended Relational Field Dynamics (HERFD) is to satisfy three demands simultaneously. The theory must be genuinely relational: no physical spacetime manifold is taken as fundamental, and no absolute spatial background is treated as ontological. It must encode a Heraclitean principle according to which exact local repetition is not generic in a truly dynamical universe. And it must remain bridgeable to the tested mathematics of general relativity and quantum theory without pretending that the bridge is merely an ordinary Lorentz transformation between two already-given spacetimes.

These demands have clear motivations. The relational demand reflects the conviction that space and time are not containers but emergent descriptions of pattern among physical degrees of freedom. This conviction has deep roots in the relational tradition running from Leibniz through Mach and into the modern shape-dynamics program of Barbour, Bertotti, Gomes, Gryb, and Koslowski [1, 2, 3]. Recent work in pure shape dynamics has further shown that the emergent arrow of time and the growth of structural complexity can be understood within a fully relational framework without appeal to external temporal structure [4]. The problem of time in canonical quantum gravity, surveyed in depth by Isham [31], makes plain that any background-independent approach must grapple with the question of how a meaningful temporal order can arise from a theory that does not presuppose it. HERFD addresses this through its Jacobi-type action and the reconstruction of emergent time from accumulated change. The canonical ADM framework [29], which reformulates general relativity in terms of spatial geometry evolving on a foliation, provides useful geometric vocabulary for the coarse-grained sector, even though HERFD's microscopic starting point is not a foliation of a pre-existing spacetime.

The Heraclitean demand reflects the idea that a genuinely historical universe should not be built from perfectly interchangeable local neighborhoods except perhaps on non-generic or unstable configurations. The bridgeability demand is methodological: a microscopic relational theory must make controlled contact with tested physics rather than remain a free-floating ontological slogan.

The ontological starting point is filamentary. The primitive entities of HERFD are extended relational filaments carrying internal fields, director frames, and interaction channels. The choice of filaments rather than point particles is deliberate. Kirchhoff rod theory and the more general Cosserat rod framework describe slender extended bodies by a centerline together with an attached frame and local strain variables for bending, twist, shear, and stretch [25, 26, 27]. These are precisely the kinds of variables needed for a relational theory of interacting extended carriers.

At the same time, the rod and linking mathematics are easiest to write by using representatives in an auxiliary oriented similarity arena $\mathcal{A} \cong \mathbb{R}^3$. This arena is not

fundamental spacetime. It is bookkeeping for filament shape, director transport, and linking data. Physical states are not raw embeddings into $\mathcal{A}$; they are equivalence classes

Andreu — HERFD

April 2026

under global similarities, filament reparametrizations, and graph automorphisms. Making this auxiliary role explicit removes any hidden-background ambiguity while preserving the filament mathematics that motivates the theory.

A direct use of absolute distance data inside the microscopic kernel would violate compatibility with the theory's own global similarity redundancy, since absolute length is gauge before scale-fixing. The present formulation therefore replaces bare distance entries by normalized, scale-covariant relational observables built from local neighborhood averages. The relational correlation length ℓ_{corr} is defined as the inverse square-root of the spectral gap of the relational graph Laplacian — a genuine HERFD-internal relational invariant. It is dimensionless at the microscopic level and acquires the dimension of length only through three emergent quantities in the Planck-matched regime: the collective action quantum $\hbar_{\text{eff}}$ (action scale), the emergent cone speed c_{eff} (causal conversion scale), and the induced gravitational coupling G_{eff} or equivalently U_0 (gravitational scale), as explained carefully in section 2.6. The ultraviolet cutoff scale ℓ_{UV} , at which the continuum description breaks down, is introduced as a conceptually distinct quantity; the identification $\ell_{\text{UV}} \sim \ell_{\text{corr}}$ is stated explicitly as a matching condition holding only in the Planck-matched regime (see section 11.8).

The topological dimension of the program is equally important. Once the basic entities are filaments, linking and entanglement become first-class relational data. The Gauss linking integral [28] provides a natural invariant for paired closed filament configurations, making topological coupling intrinsic rather than an afterthought. The integers m_{ij}^* in the linking potential are reclassified in the present paper as topological superselection sector labels, in precise analogy with the θ -vacuum structure in non-Abelian gauge theories.

The route from microscopic filament histories to effective spacetime proceeds in four stages. First, one linearizes about metastable histories and identifies regimes in which the principal symbol of the fluctuation operator is strictly hyperbolic and governed by a single characteristic cone. In the explicit bosonic tetrahedral-lattice sector, the emergent cone speed c_{eff} is derived from the ratio $\sqrt{K_{ij}/I_c}$ of continuum-limit coefficients, which is physically meaningful and independent of the lattice regulator in the properly stated continuum limit. Second, one constructs bridge maps from coarse-grained histories to effective conformal, frame, and connection data, following the logic of the Ehlers–Pirani–Schild reconstruction [10, 11, 12]. The conformal normalization $\Omega^2(F) = U(F)/U_0$ is derived as the unique factor consistent with background flatness and Einstein-frame re-

duction at background order. Third, the effective gravitational field equation is derived from a coarse-grained action principle, in the spirit of Sakharov's induced-gravity program, Jacobson's thermodynamic derivation [13], and Padmanabhan's broader perspective [21]. The induced-gravity coefficient U_0 is derived by a one-loop Seeley–DeWitt heat-kernel calculation, framed as a self-consistency relation. Fourth, one quantizes the collective sector around metastable histories, yielding a genuine bridge to effective quantum theory.

Two methodological cautions shape the development. Hotta, Murk, and Terno demon-

strate that classical–quantum hybrid systems can satisfy reasonable consistency requirements while breaking the usual symmetry–conservation link [8]. Ellis et al. provide a concrete example showing that Planck-suppressed dimension-5 operators can have significant low-energy phenomenological consequences [9], supporting the general principle that suppressed operators should not be dismissed without analysis. These observations support a restrained reading of HERFD: it should be treated as a candidate microscopic theory whose low-energy imprint is organized by controlled effective operators and collective sectors.

The paper is structured as follows. Section 2 states the five conceptual postulates, defines ℓ_{corr} with precise unit-fixing protocol, introduces ℓ_{UV} as a distinct UV scale, and classifies all dimensionless scale parameters. Section 3 formulates the microscopic configuration space with scale-covariant normalized relational observables. Section 4 introduces Heraclitean fingerprints, clarifying the hybrid local-global character of the fingerprint descriptor. Section 5 presents the core action, equations of motion, and the topological superselection reclassification. Section 6 derives emergent causal structure, the conformal normalization theorem at background order, charts, frames, and the effective gravitational action with on-shell conservation. Sections 7 and 8 construct and bridge the 2+1 lattice model. Sections 9 and 10 treat the 3+1 sector with the corrected continuum limit, hyperbolicity proof, matter-sector derivations, and chart legs. Section 11 develops the quantum bridge, gauge-invariance of the action quantum, the thermodynamic universality conjecture with explicit caveats, and the one-loop self-consistency derivation of U_0 . Section 12 explains the filament choice. Section 13 states limitations. Section 14 concludes. Appendix A collects the principal formulas.

Status Note. Throughout this paper the author distinguishes between source-based background motivations and proposed structures. The cited literature motivates rod and filament mathematics, point-separating invariant data, principal-symbol routes to emergent causal structure, relational configuration spaces, and effective spacetime recon-

struction. The specific state spaces, gauge reductions, scale-covariant relational kernels, fingerprint constructions, spectral-gap definition of ℓ_{corr} , introduction of ℓ_{UV} as a distinct UV cutoff, topological superselection reclassification, conformal normalization theorem, matter-sector derivations, one-loop self-consistency derivation of U_0 , universality proposition and conjecture, and lattice bridge maps proposed here are the author's own synthesis and programmatic proposal.

2. Conceptual Postulates of HERFD

2.1. Postulate I: Filamentary Relational Ontology

The primitive entities are extended filaments carrying internal fields, director frames, and interaction channels. No physical spacetime manifold is assumed at the microscopic level.

7

Andreu — HERFD

April 2026

What is fundamental is an unparametrized relational history of framed filaments and their interaction graph.

2.2. Postulate II: Auxiliary Similarity Arena and Gauge Reduction

A representative filament configuration may be written in an auxiliary oriented similarity arena $\mathcal{A} \cong \mathbb{R}^3$ equipped with a Euclidean metric and orientation. This arena is not ontological spacetime. It is bookkeeping for shapes, director transport, and linking data. Physical states are equivalence classes under orientation-preserving similarities, filament reparametrizations, and graph automorphisms.

2.3. Postulate III: Heraclitean Generic Non-Repeatability

Exact local repetition is not taken as a primitive prohibition imposed by fiat. Instead, the theory dynamically penalizes repeated neighborhoods and treats point-separating fingerprints as a genericity condition on physically admissible histories. This softening of the Heraclitean principle, relative to an absolute axiom, is motivated by the Manchak-Barrett hierarchy [6]: the theory prefers histories in the generic sector where local fingerprints separate neighborhoods, without requiring exact separation on every conceivable configuration.

2.4. Postulate IV: Time Emerges from Change

The physically meaningful object is an unparametrized history. Any external evolution parameter λ is gauge. Effective time is reconstructed from accumulated change through

parameter λ is gauge. Effective time is reconstructed from accumulated change through a Jacobi-type action, in the spirit of the relational dynamics of Barbour and Bertotti [1] and the scale-invariant particle dynamics of Barbour [2]. The emergent arrow of time is associated with secular structural differentiation along relational histories, as explored in shape dynamics by Barbour, Koslowski, and Mercati [4] and in the broader problem of time in quantum gravity by Isham [31].

2.5. Postulate V: Bridge to Effective Geometry and Quantum Sectors

The bridge acts on coarse-grained histories, not on a single equal-time microstate. Its first role is to produce effective conformal, frame, and connection data. Its second role, after linearization about metastable histories, is to produce a quantum theory of collective modes. At the present stage the bridge leads to effective quantum sectors, not yet to a complete microscopic derivation of all of quantum mechanics.

2.6. Internal Definition of the Relational Correlation Length and the Unit-Fixing Protocol

Because the redundancy group G includes a global similarity scaling $\mathbb{R}^+$, absolute length is gauge before unit-fixing. The correlation length of the filament network cannot be a primitive input; it must be defined from HERFD-internal relational data.

Definition 2.1 (Relational correlation length). Let q^* be a metastable background history in the generic Heraclitean sector. The relational graph Laplacian $\Delta_R[q^*]$ defined in eq. (15) below is a positive semidefinite self-adjoint operator on the space of node functions with respect to the inner product $\langle f, g \rangle = \sum_{\alpha} \rho_{\alpha} f_{\alpha} g_{\alpha}$. Its spectrum satisfies $0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \dots$, where $\lambda_0 = 0$ corresponds to the constant eigenfunction on each connected component. The *relational correlation length* of q^* is

$$\ell_{\text{corr}}[q^*] := \lambda_1(\Delta_R[q^*])^{-1/2}. \quad (1)$$

Proposition 2.2 (Properties of ℓ_{corr}). (i) **Gauge invariance.** Since the normalized separation $\hat{\sigma}_{\alpha\beta}$ defined in eq. (10) is invariant under the full redundancy group G , the eigenvalues of Δ_R are G -invariant. In particular, $\ell_{\text{corr}}[q^*] = \ell_{\text{corr}}[g \cdot q^*]$ for all $g \in G$.

$$g \in \mathcal{G}.$$

(ii) **Dimensionless at the microscopic level.** Because $\hat{\sigma}_{\alpha\beta}$ is dimensionless by construction (it is a ratio of squared distances), the kernel weights $K_{\alpha\beta}$ are dimensionless, and $\lambda_1(\Delta_R)$ is a dimensionless pure number. Consequently $\ell_{\text{corr}}[q^*]$ defined by eq. (1) is a dimensionless HERFD-internal relational quantity before unit-fixing. The unit-fixing protocol below explains how it acquires physical dimensions.

~~(iii) **Background dependence.** Different non-gauge-equivalent metastable backgrounds will generically have different spectral gaps, and hence different dimensionless correlation lengths.~~

(iv) **Spectral interpretation.** The correlation length measures the spatial scale beyond which the low-frequency modes of Δ_R are not significantly excited. The direct kernel suppression scale α_R controls the width of each pairwise kernel weight, while ℓ_{corr} characterizes the network-level spectral gap of the Laplacian built from those weights. These are distinct quantities playing distinct roles.

Unit-fixing protocol. Before unit-fixing, ℓ_{corr} is a pure number. Physical dimensions enter through three steps.

Step 1. The collective action quantum $\hbar_{\text{eff}}^-[q^*]$ (Section 11.7) carries dimensions of action $[\text{J} \cdot \text{s}]$ by the physical identification of the symplectic form with Planck's constant in the low-energy quantum sector.

Step 2. The emergent cone speed c_{eff} (section 10.2) carries dimensions of velocity $[\text{m/s}]$ by the physical identification of the null cone with the speed of light.

[↓ Download PDF](#)
[🔒 Full PDF Package](#)
[🔒 Translate](#)
Andreu — HERFD
April 2026

11.2. Canonical Momentum and Symplectic Form

The canonical momentum of the linearized sector is $\Pi_A = G_{AB}(q^*) D_\tau \xi^B$, and the natural symplectic form on the solution space is

$$\Omega(\xi_1, \xi_2) = \int_{\Sigma_\tau} (\xi_1^A \Pi_{2A} - \xi_2^A \Pi_{1A}). \quad (72)$$

Once a positive-frequency decomposition is available in the metastable sector—guaranteed in the one-cone hyperbolic regime of theorem 10.2— Ω supplies the one-particle structure required for quantization.

11.3. Finite-Dimensional Collective-Coordinate Bridge

If only finitely many slow collective coordinates Q^a survive the coarse graining, the effective quantum bridge is

$$B_Q^{\text{coll}} : [q^\star]_{\text{cg}} \mapsto (\mathcal{H}_{\text{coll}}, \hat{H}_{\text{coll}}(\tau), \hat{\rho}_{\text{coll}}), \quad (73)$$

with

$$\mathcal{H}_{\text{coll}} = L^2(\mathcal{C}_{\text{coll}}, d\mu_G), \quad \hat{H}_{\text{coll}} = -\frac{\bar{h}_{\text{eff}}[q^\star]^2}{2} \Delta_G + V_{\text{eff}}(Q, \tau), \quad (74)$$

and the emergent-time Schrödinger equation

$$i\bar{h}_{\text{eff}}[q^\star] \partial_\tau \Psi(Q, \tau) = \hat{H}_{\text{coll}} \Psi(Q, \tau). \quad (75)$$

The action quantum $\bar{h}_{\text{eff}}[q^\star]$ labels the collective sector and is defined in section 11.7.

11.4. Field-Theoretic Quantum Bridge

Working around the metastable background $q^\star$ and integrating over fluctuations:

$$Z_{\text{HERFD}}^{(q^\star)}[J] = \int \mathcal{D}\xi \exp \left(\frac{i}{\bar{h}_{\text{eff}}[q^\star]} S_{q^\star}^{(2)}[\xi] + \frac{i}{\bar{h}_{\text{eff}}[q^\star]} \int J_A \mathcal{O}^A[\xi; q^\star] \right). \quad (76)$$

In the weak-fluctuation, one-cone, long-wavelength regime, one demands the matching relation

$$Z_{\text{HERFD}}^{(q^\star)}[J] \approx Z_{\text{eff}}[g, e, \omega; J] = \int \mathcal{D}\varphi \exp \left(\frac{i}{\bar{h}_{\text{eff}}[q^\star]} S_{\text{eff}}[\varphi; g, e, \omega] + \frac{i}{\bar{h}_{\text{eff}}[q^\star]} \int J\varphi \right). \quad (77)$$

A suitable low-energy sector of HERFD thereby reproduces an effective quantum field theory on the emergent geometry, with the action scale carried by the metastable history rather than inserted as an independent microscopic postulate [7].

11.5. Conditional Spinorial Sector

A spinorial bridge is admitted only conditionally: the emergent frame bundle must lift to a spin bundle, and the microscopic orientational or topological sector must support the required $\mathbb{Z}_2$ double cover. Until that microscopic derivation is supplied, the spinor leg remains labeled conditional.

11.6. Honest Claim Boundary

The theory supports the following concrete claim levels: a concrete hyperbolicity theorem for the bosonic tetrahedral sector with the emergent cone speed derived as a regulator-independent quantity (theorem 10.2); a classical bridge to conformal, frame, and connection data proved as a continuum-limit theorem (theorem 10.5 and corollary 10.6); the conformal normalization $\Omega^2(F) = U(F)/U_0$ established as a unique structural identification at background order (proposition 6.2); the matter-sector functions $Z(F)$ and $M_\varphi^2(F)$ expressed in terms of $U(F)$ and determined at background order (corollary 10.7); an effective gravitational field equation derived from a coarse-grained action with on-shell covariant conservation (proposition 6.6); a quantum bridge for collective coordinates and low-energy bosonic fluctuation fields, with the action quantum proved gauge-invariant within gauge-equivalence classes (proposition 11.1); and the induced-gravity coefficient U_0 derived by a one-loop Seeley–DeWitt heat-kernel self-consistency calculation (section 11.8). The conditional route to spinorial sectors, derivation of the Born rule, universality of h_{eff}^- across all metastable histories, and the full functional form of $U(F)$ beyond U_0 remain open.

11.7. Emergent Quantum of Action and Gauge-Invariance

The linearized theory supplies the needed kinematic ingredients. Let Q^a denote slow collective coordinates, P_a their canonically conjugate momenta, and $\Theta_{\text{coll}} = P_a dQ^a$ the collective canonical one-form. For any primitive closed cycle $\mathcal{C}$ in the metastable collective sector, define its positive action content by

$$A[\mathcal{C}] = \left| \oint_{\mathcal{C}} \Theta_{\text{coll}} \right|. \quad (78)$$

Impose the metastable spectral-gap hypothesis: the set of positive action values is discrete near the origin and possesses a smallest strictly positive element. The emergent quantum of action is then:

$$h_{\text{eff}}[q^*] := \min\{A[\mathcal{C}] > 0\}, \quad h_{\text{eff}}^-[q^*] := \frac{h_{\text{eff}}[q^*]}{2\pi}. \quad (79)$$

Proposition 11.1 (Gauge-equivalence class universality of h_{eff}). *Let $q_2^* = g \cdot q_1^*$ for some $g \in G$. Then $h_{\text{eff}}[q_1^*] = h_{\text{eff}}[q_2^*]$.*

Proof. The collective canonical one-form $\Theta_{\text{coll}} = r_a a \dot{q}^a$ is defined on the reduced phase space $T^*\mathcal{Q}_{\text{HERFD}}$ obtained after quotienting by G . Since $\mathcal{Q}_{\text{HERFD}} = \mathcal{Q}_{\text{kin}}/G$, the reduced coordinates (Q^a, P_a) are G -invariant by construction, and Θ_{coll} is a well-defined G -invariant one-form. For any primitive closed cycle $\mathcal{C}$ based at q_1^* : $\oint_{g \cdot \mathcal{C}} \Theta_{\text{coll}} = \oint_{\mathcal{C}} g^* \Theta_{\text{coll}} = \oint_{\mathcal{C}} \Theta_{\text{coll}}$, so $A[g \cdot \mathcal{C}] = A[\mathcal{C}]$, and taking the minimum gives $h_{\text{eff}}[q_2^*] = h_{\text{eff}}[q_1^*]$. $\square$

Remark 11.2. Proposition 11.1 establishes that h_{eff} is well-defined on the reduced configuration space $\mathcal{Q}_{\text{HERFD}}$. Whether physically distinct (non-gauge-equivalent) metastable histories give the same h_{eff} requires a universality theorem beyond what is proved here.

Conjecture 11.3 (Thermodynamic universality of h_{eff} in the bosonic tetrahedral sector). *In the bosonic tetrahedral sector with periodic boundary conditions on T_L^3 , the minimum primitive-cycle action at a background q^* satisfying assumption 10.1 and the metastable spectral-gap hypothesis is:*

$$h_{\text{eff}}[q^*] = 2\pi \frac{c_{\text{eff}}[q^*]}{L} \cdot G_{11}(q^*), \quad (80)$$

where $c_{\text{eff}}[q^*]$ is the emergent cone speed eq. (63), L is the system size, and $G_{11}(q^*)$ is the inertial weight of the minimal collective mode. To give a HERFD-internal determination of L , one may consider the cosmological specialization of the effective field equation eq.(41) to a homogeneous and isotropic (FRW) reduction. Such a reduction has not been derived from HERFD in the present paper; it is introduced here as a motivating example only. Under an assumed FRW reduction with $\overline{W_0}/(3U_0)$ playing the role of the cosmological constant, the system scale would be $L = H^{-1} = \sqrt{3U_0/W_0}$. If c_{eff} , U_0 , W_0 , and G_{11} are universal constants of a given symmetry class of metastable histories, then h_{eff} would be universal within the class. Confirming this conjecture requires a HERFD renormalization group analysis and a derivation of the FRW reduction, both of which are beyond the scope of the present paper.

11.8. One-Loop Self-Consistency Derivation of the Induced-Gravity Coefficient U_0

The effective coarse-grained action eq. (40) postulates an induced-gravity coupling $U(F)$ whose background value $U_0 = U(F_0)$ determines the effective Newton constant. In this section we derive a self-consistency relation for U_0 from the HERFD fluctuation spectrum by a one-loop heat-kernel calculation, following the induced-gravity tradition of Sakharov and Visser [14].

Conceptual status. The one-loop result derived below is *not* an independent first-principles determination of U_0 from a pre-fixed UV scale. Rather, it is a self-consistency relation: the UV cutoff scale ℓ_{UV} (distinct from ℓ_{corr} ; see below) enters the calculation as an input, and the result determines U_0 given the physical identification of ℓ_{UV} . The

identification $\ell_{UV} \sim \ell_{\text{corr}}$ is then imposed as a *matching condition* in the Planck-matched regime:

$$\ell_{UV} = \beta_{UV} \ell_{\text{corr}} \quad (81)$$

for some dimensionless matching factor β_{UV} of order unity.

To maintain compact notation throughout the one-loop formulas, we absorb β_{UV} into the scheme factor $f(\alpha_R^*)$ by defining the *total scheme factor*

$$f_{\text{tot}}(\alpha_R^*, \beta_{UV}) := \frac{f(\alpha_R^*)}{\beta_{UV}^2}. \quad (82)$$

All subsequent one-loop formulas are written in terms of f_{tot} . This keeps the displayed equations compact while making the $O(1)$ matching ambiguity in β_{UV} fully explicit through this single definition. The identification $\ell_{UV} \sim \ell_{\text{corr}}$ is a physically motivated assumption in the Planck-matched regime, not a consequence of the spectral definition eq. (1). The separation between ℓ_{UV} and ℓ_{corr} reflects their distinct roles: ℓ_{corr} characterizes network-level spectral correlations, while ℓ_{UV} is the scale at which the continuum field-theoretic description breaks down.

Setup

Restricting to the bosonic tetrahedral sector, the collective-mode fluctuation operator in the emergent spacetime description (after canonical rescaling to $\tilde{\theta} = I_c^{1/2} \theta$) is the standard minimal scalar operator $\mathcal{L} = -\square_g + M^2$ for the flat emergent metric in the background-flatness limit.

One-loop effective action

Integrating out N_θ bosonic filament modes gives the one-loop effective action:

$$S^{(1\text{-loop})}[q^*] = \frac{i\hbar_{\text{eff}}}{2} N_\theta \text{Tr} \log \mathcal{L}. \quad (83)$$

Using the heat-kernel representation $\text{Tr} \log \mathcal{L} = -\int_0^\infty (ds/s) K(s)$ with $K(s) = \text{Tr} e^{-s\mathcal{L}}$, and the Seeley–DeWitt asymptotic expansion for a second-order minimal operator in 3+1 dimensions:

$$K(s) \sim \frac{1}{(4\pi s)^2} \int d^4x \sqrt{-g} \left(1 + \frac{1}{6} R[\tilde{g}] s + O(s^2) \right) \quad \text{as } s \rightarrow 0^+. \quad (84)$$

Andreu — HERFD

April 2026

Einstein–Hilbert coefficient

The term proportional to R in the one-loop effective action integrates over s from the UV cutoff $1/\Lambda_{\text{UV}}^2$ to the IR scale $1/\mu^2$:

$$S^{(1\text{-loop})} \supset \frac{i\bar{h}_{\text{eff}} N_\theta}{192\pi^2} \int d^4x \sqrt{-g} R \Lambda_{\text{UV}}^2. \quad (85)$$

UV cutoff identification and matching condition

The natural UV cutoff in HERFD is the inverse UV scale $\Lambda_{\text{UV}} = \ell_{\text{UV}}^{-1}$, where ℓ_{UV} is the scale at which the continuum description breaks down—the scale below which modes are already integrated out in passing from the microscopic HERFD action to the quadratic collective action $S^{(2)}$. This scale is conceptually distinct from ℓ_{corr} .

In the Planck-matched regime, the matching condition eq. (81) is imposed, so that $\ell_{\text{UV}} = \beta_{\text{UV}} \ell_{\text{corr}}$. Substituting $\Lambda_{\text{UV}} = (\beta_{\text{UV}} \ell_{\text{corr}})^{-1}$ into the one-loop result eq. (85):

$$S^{(1\text{-loop})} \supset \frac{i\bar{h}_{\text{eff}} N_\theta}{192\pi^2 \beta_{\text{UV}}^2 \ell_{\text{corr}}^2} \int d^4x \sqrt{-g} R. \quad (86)$$

Defining f_{tot} as in eq. (82) absorbs the factor β_{UV}^{-2} together with the regularization scheme factor $f(\alpha_R^*)$ into a single $O(1)$ coefficient.

Self-consistency relation for U_0

Comparing with the Einstein–Hilbert form $\int d^4x \sqrt{-g} U_0 R$, the one-loop calculation yields the self-consistency relation:

$$U_0 = \frac{N_\theta \bar{h}_{\text{eff}}}{192\pi^2 \ell_{\text{corr}}^2} \cdot f_{\text{tot}}(\alpha_R^*, \beta_{\text{UV}}), \quad (87)$$

where $f_{\text{tot}}(\alpha_R^*, \beta_{\text{UV}}) = f(\alpha_R^*)/\beta_{\text{UV}}^2$ is the total scheme factor defined in eq. (82), dimensionless and $O(1)$. This result should be interpreted as follows: given the UV identification $\ell_{\text{UV}} = \beta_{\text{UV}} \ell_{\text{corr}}$ in the Planck-matched regime, the one-loop consistency equation determines U_0 in terms of N_θ , $\bar{h}_{\text{eff}}$, ℓ_{corr} , and the combined factor f_{tot} . It is not an independent derivation of U_0 from a separately fixed cutoff.

Consequences and the species consistency relation

The effective Planck length from eq. (64) becomes:

$$\ell_{P,\text{eff}} = \sqrt{\frac{h_{\text{eff}}^-}{16\pi U_0}} = \ell_{\text{corr}} \sqrt{\frac{12\pi}{N_\theta f_{\text{tot}}(\alpha_R^*, \beta_{\text{UV}})}}. \quad (88)$$

32

Andreu — HERFD

April 2026

The Planck-matched regime requires $\ell_{P,\text{eff}} = \ell_{\text{corr}}$, giving the leading-order species consistency relation:

$$N_\theta = \frac{12\pi}{f_{\text{tot}}(\alpha_R^*, \beta_{\text{UV}})} \approx \frac{37.7}{f_{\text{tot}}}. \quad (89)$$

This should be understood as a *leading-order species consistency relation* implied by the current one-loop truncation and the UV identification $\ell_{\text{UV}} = \beta_{\text{UV}} \ell_{\text{corr}}$, not as a firm HERFD prediction. Its epistemic status is that of a one-loop matching condition: for the Planck scale to coincide with the HERFD correlation scale under the matching condition eq. (81), the bosonic sector must contain approximately $12\pi/f_{\text{tot}}$ degrees of freedom at leading order in the one-loop approximation.

The effective Newton constant is:

$$G_{\text{eff}} = \frac{c_{\text{eff}}^3}{16\pi U_0} = \frac{12\pi c_{\text{eff}}^3 \ell_{\text{corr}}^2}{N_\theta h_{\text{eff}}^- f_{\text{tot}}(\alpha_R^*, \beta_{\text{UV}})}, \quad (90)$$

and the cosmological constant problem is translated into the HERFD internal condition $W_0 \ell_{\text{corr}}^2 / U_0 = 192\pi^2 \mu_H \rho_0^2 \ell_{\text{corr}}^5 / (N_\theta h_{\text{eff}}^- f_{\text{tot}}) \ll 1$.

Scope and limitations of the one-loop calculation

The derivation of eq. (87) involves three approximations. First, a *one-loop truncation*: higher-loop contributions involve additional powers of the interaction couplings λ_H , μ_H , and rod-sector constants. These contributions are expected to be subleading for sufficiently many bosonic species, though a systematic large- N_θ expansion with the combinatorial counting that would justify $1/N_\theta$ as the loop parameter has not been developed in this paper. Second, a *minimal coupling approximation*: the bosonic field is treated as minimally coupled to the emergent metric in the background-flatness limit. Non-minimal fingerprint-coupling corrections enter at order $(\nabla F)^2 / \ell_{\text{corr}}^2$ and contribute to the fingerprint-dependent correction $U(F)$ beyond U_0 . Third, the *UV matching condition* $\ell_{\text{UV}} = \beta_{\text{UV}} \ell_{\text{corr}}$: this is a physically motivated identification in the Planck-matched regime, but it introduces an $O(1)$ uncertainty through the combined factor $f_{\text{tot}}(\alpha_R^*, \beta_{\text{UV}}) =$

$f(\alpha_R^*)/\beta_{UV}^2$, which has been made fully explicit through eq. (82).

12. Why Filaments

The choice of filaments as primitive carriers is not incidental. The mathematics of HERFD relies throughout on centerlines, director fields, curvature and twist variables, linking integrals, and coarse-grained lattice networks of extended carriers. This is the mathematics of filaments and rods, not the mathematics of perturbative strings on a prescribed target manifold, nor the mathematics of spin networks on a fixed combinatorial complex. The Cosserat rod variables—the director triad, the curvature and twist, the shear and stretch—are not artificially introduced; they are precisely what is needed to

describe shape, orientation, and deformation of extended one-dimensional carriers in a representation-independent way.

The centerline and director data are written in representatives that live in an auxiliary similarity arena. This does not weaken the filament choice; it makes the kinematics honest. The rod variables and linking integrals still do the conceptual work, but they are clearly treated as representative data before gauge reduction rather than as hidden coordinates on a fundamental background spacetime. The analogy with the DeWitt supermetric [2], where the space of spatial geometries is quotiented by diffeomorphisms to obtain the superspace of physical geometries, is direct and deliberate.

The distinction from string theory deserves emphasis. Perturbative string theory begins with an action principle on a prescribed target spacetime manifold, and the string is a one-dimensional object whose vibration modes produce particle-like states in that target spacetime. HERFD instead treats filaments as the ontological primitives and constructs spacetime as an emergent, coarse-grained description of filament configurations. There is no target manifold in the ontology; the similarity arena is bookkeeping, not a stage. The reference to recent work on emergent light cones in relational shape dynamics [5] further motivates the plausibility of this route.

13. Limitations and Methodological Cautions

The following items have not been established in the present paper and are listed here for completeness and transparency.

1. **Genericity of the one-cone regime.** Theorem 10.2 covers the explicit bosonic tetrahedral sector only. A proof that the one-cone regime is generic for all admissible

metastable filament histories remains open.

2. **Full derivation of 3+1 Einstein gravity.** The effective action eq. (40) is an ansatz for the coarse-grained sector, not a derived consequence of the microscopic dynamics, except in that U_0 is now computed to one-loop order under the UV matching condition eq. (81).
3. **Standard Model gauge sector and fermionic matter.** A derivation of the ~~Standard Model gauge sector or of fermionic matter from first principles has not~~ been undertaken.
4. **Complete quantum theory.** A complete quantum theory of the microscopic filaments has not been developed.
5. **Conditional spinorial sector.** The spinorial bridge remains conditional on the microscopic frame bundle admitting a spin structure.

6. **Heraclitean metric separation.** A proof that every effective metric arising in the theory is Heraclitean in the exact Manchak–Barrett sense [6] has not been given.
7. **Full functional form of $U(F)$.** Section 11.8 derives only $U_0 = U(F_0)$. The fingerprint-gradient corrections to $U(F)$ are stated to require a higher-order calculation that has not been performed.
8. **Full matter-sector functions.** Corollary 10.7 reduces $Z(F)$ and $M_\varphi^2(F)$ to $U(F)$, which is determined only at background order.
9. **Universality of h_{eff}^- .** Proposition 11.1 establishes gauge-equivalence class universality only. Conjecture 11.3 provides a motivating thermodynamic scenario that depends on a FRW cosmological specialization of eq. (41) that has not been derived from HERFD.
10. **Self-consistency of the one-loop U_0 .** The derivation of U_0 in section 11.8 is a self-consistency relation, not an independent derivation, and rests on the UV matching condition $\ell_{\text{UV}} = \beta_{\text{UV}} \ell_{\text{corr}}$. The $O(1)$ matching ambiguity is absorbed into the total scheme factor $f_{\text{tot}}(\alpha_R^*, \beta_{\text{UV}}) = f(\alpha_R^*)/\beta_{\text{UV}}^2$ as defined in eq. (82). A systematic large- N_θ expansion justifying the loop-suppression claim has not been developed. The species consistency relation eq. (89) is a leading-order consistency relation, not

a firm prediction, and its epistemic status should be understood accordingly.

11. **Analytic determination of α_R^* and $\alpha_H^{\max}$.** A full analytic determination of α_R^* and $\alpha_H^{\max}$ in the tetrahedral sector remains to be carried out numerically.
12. **Coupling constants as model parameters.** The action-level coupling constants $\lambda_H, \eta_H, B_i^{ab}, S_i^{ab}, C_i, D_i$, the supermetric coefficients, and the linking-sector couplings Λ_{ij}, μ_L are constrained by stability (assumption 5.1) and cosmological bounds, but are not derived from a deeper HERFD micro-law. They are honest model parameters analogous to coupling constants in the Standard Model, whose determination would require a UV completion or experimental matching. The model-specific result of Ellis et al. [9] in the context of minimal supersymmetric SU(5) supports the general principle that suppressed operators should not be dismissed without analysis.

The Heraclitean condition is interpreted as a genericity requirement: the theory dynamically prefers point-separating fingerprints, but exact separation is treated as a generic sector rather than an unconditional axiom. The Hotta–Murk–Terno result [8] warns that introducing open or hybrid effective sectors may break conservation; HERFD therefore treats conservation as exact only in fully reversible sectors derivable from the variational principle eq. (40).

14. Conclusion

This paper has presented a full formulation of Heraclitean Extended Relational Field Dynamics, a framework in which the primitive ontology consists of extended relational filaments whose natural mathematics is closer to Kirchhoff–Cosserat rod theory, framed curves, and linking integrals than to any perturbative string formalism or spin-network combinatorics.

The development has proceeded in several stages. The microscopic state space has been written in filament language with an explicit auxiliary similarity arena and gauge reduction. The relational correlation length ℓ_{corr} has been defined internally as the inverse square-root of the spectral gap of the relational graph Laplacian (definition 2.1), with a careful unit-fixing protocol that resolves the tension between its dimensionless microscopic status and its use as a physical length after Planck matching (proposition 2.2 and the unit-fixing protocol in section 2.6). A conceptually distinct UV cutoff scale ℓ_{UV} has been introduced, with the identification $\ell_{\text{corr}} \sim \ell_{\text{UV}}$ stated as a matching condition in the

been introduced, with the identification $\kappa_{UV} = \kappa_{\text{eff}}$ stated as a matching condition in the Planck-matched regime rather than a definitional equivalence (eq. (81)). All four dimensionless α -parameters have been classified with precision (proposition 2.4): α_a is a UV regulator with physical content entering only through K_{ij} and I_c (defined in eq. (56)) held fixed in the continuum limit; α_R is fixed by RG optimization; α_H is bounded by chart admissibility; and α_{ij} is irrelevant in the smooth regime. The linking integers m_{ij}^* have been reclassified as topological superselection sector labels (proposition 5.2). The fingerprint field's hybrid local-global character has been explicitly acknowledged (section 4.3).

The continuum limit of the bosonic tetrahedral sector has been stated with the correct scaling of bond stiffness $\kappa^A \propto 1/a^2$ (assumption 10.1), resolving the tension between the continuum limit and the emergence of a finite physical speed. The emergent cone speed $c_{\text{eff}} = (2/\sqrt{3})\sqrt{\kappa}\gamma I$ has been shown to be independent of the lattice regulator a , and the Planck-length formula has been shown to involve c_{eff}^3 cancelling exactly (remark 10.3). The conformal normalization $\Omega^2(F) = U(F)/U_0$ has been derived at background order as a unique structural identification (proposition 6.2), with explicit acknowledgement that: the background limit yields an Einstein equation with an effective cosmological term W_0 , not the sourceless Einstein equation (eq. (34)); and the full function $U(F)$ beyond U_0 is not derived in this paper. The effective Planck length is defined by three emergent quantities $\bar{h}_{\text{eff}}$, c_{eff} , and G_{eff} (equivalently U_0), all required as independent inputs (eq. (2)). The matter-sector parameters $Z(F)$ and $M_\varphi^2(F)$ have been expressed in terms of $U(F)$ and determined at background order (corollary 10.7). The effective gravitational field equation has been derived from a coarse-grained action with on-shell covariant conservation (proposition 6.6). The action quantum $\bar{h}_{\text{eff}}[q^*]$ has been proved gauge-invariant within gauge-equivalence classes (proposition 11.1). The thermodynamic universality conjecture (conjecture 11.3) has been stated with explicit acknowledgement that it depends on a FRW reduction not derived in this paper. The

induced-gravity coefficient U_0 has been derived by a one-loop Seeley–DeWitt heat-kernel self-consistency calculation (section 11.8), with the matching ambiguity β_{UV} absorbed into the total scheme factor $f_{\text{tot}}(\alpha_R^*, \beta_{UV}) = f(\alpha_R^*)/\beta_{UV}^2$ as defined in eq. (82), making the $O(1)$ uncertainty fully explicit. The species consistency relation $N_\theta \approx 12\pi/f_{\text{tot}}$ is classified as a leading-order consistency relation rather than a firm prediction.

The core conceptual structure of the theory is:

filament history
 $\longrightarrow$ fingerprint chart + principal symbol

$$\begin{aligned}
&\longrightarrow \text{conformal class} \times \Omega^2(F_0) + \text{frame} + \text{connection} \\
&\longrightarrow \text{effective GR/QM sectors} + \{c_{\text{eff}}, \bar{h}_{\text{eff}}, U_0\}_{\text{HERFD-internal}}.
\end{aligned} \tag{91}$$

This structure preserves the relational premise, treats the Heraclitean condition as a serious mathematical requirement without over-rigidity, makes curvature an emergent property of filament dynamics, and provides a defensible bridge—with concrete theorems in the bosonic sector—to effective geometry and controlled quantum sectors, without collapsing the microscopic theory into standard general relativity or quantum mechanics.

A. Compact List of Principal Formulas

Reduced state space:

$$\mathcal{Q}_{\text{HERFD}} = \frac{\prod_i (\text{Emb}(\Sigma_i, \mathcal{A}) \times \text{Map}(\Sigma_i, SO(3)) \times \mathcal{F}_i) \times \Gamma}{\text{Sim}^+(3) \times \prod_i \text{Diff}(\Sigma_i) \times \text{Aut}(\Gamma)}. \tag{A.1}$$

Relational correlation length:

$$\ell_{\text{corr}}[q^*] = \lambda_1(\Delta_R[q^*])^{-1/2} \quad (\text{dimensionless before unit-fixing}). \tag{A.2}$$

UV cutoff scale, matching condition, and total scheme factor:

$$\begin{aligned}
&\ell_{\text{UV}} \text{ (distinct from } \ell_{\text{corr}}); \quad \ell_{\text{UV}} = \beta_{\text{UV}} \ell_{\text{corr}} \text{ in the Planck-matched regime;} \\
&f_{\text{tot}}(\alpha_R^*, \beta_{\text{UV}}) := \frac{f(\alpha_R^*)}{\beta_{\text{UV}}^2} \text{ (absorbs matching ambiguity into scheme factor)}.
\end{aligned} \tag{A.3}$$

Topological superselection decomposition:

$$\mathcal{Q}_{\text{HERFD}} = \bigsqcup_{m \in \mathbb{Z}^{|\Gamma_{\text{cl}}|}} \mathcal{Q}_{\text{HERFD}}^{(m)}. \tag{A.4}$$

Normalized relational kernel:

$$\hat{R}_{\alpha\beta} = (\hat{\sigma}_{\alpha\beta}, t_i \cdot t_j, t_i \cdot \hat{n}_{\alpha\beta}, t_j \cdot \hat{n}_{\alpha\beta}, Q_{\alpha\beta}, \theta_i - \theta_j, \langle \Phi_i, \Phi_j \rangle), \quad \hat{\sigma}_{\alpha\beta} = \frac{|\Delta_{\alpha\beta}|^2}{d_\alpha d_\beta}. \tag{A.5}$$

Bond stiffness scaling in the continuum limit:

$$\kappa_n^A = \kappa_n^{\mathcal{A}}/a^2 \quad \text{with } \kappa \text{ bounded uniformly in } a. \quad (\text{A.6})$$

Continuum-limit tensor fields (3+1 sector):

$$K_{ij}(X) = \frac{1}{\Omega_c} \sum_{A=1}^4 \kappa^A(X) \ell_A^i \ell_A^j, \quad I_c(X) = \frac{I(X)}{\Omega_c}, \quad M^2(X) = \frac{M_n^2(X)}{\Omega_c}. \quad (\text{A.7})$$

Jacobi action and emergent time:

$$S[q] = \int d\lambda \sqrt{2(E - V[q])} \mathcal{G}_{IJ}(q) \dot{q}^I \dot{q}^J, \quad d\tau^2 = \frac{\mathcal{G}_{IJ}(q) dq^I dq^J}{2(E - V[q])}. \quad (\text{A.8})$$

Hamiltonian constraint:

$$H(q, p) = \frac{1}{2} \mathcal{G}^{IJ}(q) p_I p_J + V[q] - E = 0. \quad (\text{A.9})$$

Frozen-coefficient symbol (bosonic tetrahedral sector, stiffness scaling applied):

$$p_a(\omega, \xi; X_0) = I_c(X_0) \omega^2 - K_{ij}(X_0) \xi^i \xi^j - M^2(X_0) + O(a^2 |\xi|^4). \quad (\text{A.10})$$

Emergent speed of light (regulator-independent):

$$c_{\text{eff}} = \sqrt{\left. \frac{K_{ij} \hat{\xi}^i \hat{\xi}^j}{I_c} \right|_{\text{iso}}} = \frac{2}{\sqrt{3}} \sqrt{\kappa^{\mathcal{A}}}. \quad (\text{A.11})$$

Effective Planck length (three independent inputs required):

$$\ell_{P,\text{eff}} = \left(\frac{\bar{h}_{\text{eff}} G_{\text{eff}}}{c_{\text{eff}}^3} \right)^{1/2} = \sqrt{\frac{\bar{h}_{\text{eff}}}{16\pi U_0}}. \quad (\text{A.12})$$

Conformal normalization at background order:

$$\Omega^2(F) = \frac{U(F)}{U_0}, \quad U_0 = U(F_0) \text{ (derived to one-loop; full } U(F) \text{ open)}. \quad (\text{A.13})$$

Background field equation (Einstein with effective cosmological term):

$$U_0 G_{\mu\nu} = T_{\mu\nu}^{(\varphi)} - g_{\mu\nu} W_0. \quad (\text{A.14})$$

Matter-sector parameters (at background order):

$$Z(F)|_{F_0} = 1, \quad M_\varphi^2|_{F_0} = \frac{M^2}{I_c}. \quad (\text{A.15})$$

Effective coarse-grained action:

$$S_{\text{eff}}[g, F, \varphi] = \int d^4x \sqrt{-g} \left[U(F) R - \frac{\lambda_H \eta_H}{2} G_{AB} \nabla^\mu F^A \nabla_\mu F^B - W(F) + \mathcal{L}_\varphi \right]. \quad (\text{A.16})$$

Effective gravitational field equation:

$$U(F) G_{\mu\nu} + (g_{\mu\nu} \square_g - \nabla_\mu \nabla_\nu) U(F) = T_{\mu\nu}^{(\varphi)} + \lambda_H \Xi_{\mu\nu} - g_{\mu\nu} W(F). \quad (\text{A.17})$$

One-loop self-consistency relation for U_0 (given $\ell_{\text{UV}} = \beta_{\text{UV}} \ell_{\text{corr}}$):

$$U_0 = \frac{N_\theta h_{\text{eff}}^-}{192\pi^2 \ell_{\text{corr}}^2} \cdot f_{\text{tot}}(\alpha_R^*, \beta_{\text{UV}}), \quad f_{\text{tot}} = f(\alpha_R^*) / \beta_{\text{UV}}^2. \quad (\text{A.18})$$

Leading-order species consistency relation (one-loop, $\ell_{\text{UV}} = \beta_{\text{UV}} \ell_{\text{corr}}$):

$$N_\theta = \frac{12\pi}{f_{\text{tot}}(\alpha_R^*, \beta_{\text{UV}})} \approx \frac{37.7}{f_{\text{tot}}} \quad (\text{Planck-matched regime, one-loop truncation}). \quad (\text{A.19})$$

Emergent action quantum and effective Planck length:

$$h_{\text{eff}}[q^*] = \min\{A[\mathcal{C}] > 0\}, \quad h_{\text{eff}}^-[q^*] = \frac{h_{\text{eff}}[q^*]}{2\pi}. \quad (\text{A.20})$$

Classical bridge outputs:

$$B_{2+1}[\gamma_{2+1}] = (\mathcal{M}_{2+1}, [g_{\mu\nu}], e^a{}_\mu, \omega_{ab\mu}, \varphi_{2+1}, \Xi_{\mu\nu}). \quad (\text{A.21})$$

$$B_{\text{class}}^{(3+1)}[\gamma] = (\mathcal{M}_{3+1}, [g], e, \omega, \varphi_{\text{eff}}, U(F), \Xi_{\mu\nu}). \quad (\text{A.22})$$

Collective quantum bridge:

$$B_Q^{\text{coll}} : [q^*]_{\text{cg}} \mapsto (\mathcal{H}_{\text{coll}}, \hat{H}_{\text{coll}}(\tau), \rho_{\text{coll}}), \quad i h_{\text{eff}}^-[q^*] \partial_\tau \Psi = \hat{H}_{\text{coll}} \Psi. \quad (\text{A.23})$$

References

- [1] J. Barbour and B. Bertotti, *Mach's principle and the structure of dynamical theories*, Proc. R. Soc. Lond. A **382** (1982), 295–306.
- [2] J. Barbour, *Scale-invariant gravity: Particle dynamics*, Class. Quantum Grav. **20** (2003), 1543–1570, arXiv:gr-qc/0211021.
- [3] H. Gomes, S. Gryb, and T. Koslowski, *Einstein gravity as a 3D conformally invariant theory*, Class. Quantum Grav. **28** (2011), 045005, arXiv:1010.2481 [gr-qc].
- [4] J. Barbour, T. Koslowski, and F. Mercati, *Identification of a gravitational arrow of time*, Phys. Rev. Lett. **113** (2014), 181101, arXiv:1409.0917 [gr-qc].
- [5] F. S. N. Lobo, *On the possibility of emergent light cones from relational shape dynamics*, arXiv:2602.23853 [gr-qc] (2026).
- [6] J. B. Manchak and T. Barrett, *A hierarchy of spacetime symmetries: Holes to Heraclitus*, forthcoming in The British Journal for the Philosophy of Science, preprint (2023).
- [7] M. Baldo, *Lorentz transformation in quantum mechanics*, arXiv:2511.11342 [quant-ph] (2025).
- [8] M. Hotta, S. Murk, and D. R. Terno, *Classical–quantum systems breaking conservation laws*, arXiv:2506.15291 [quant-ph] (2025).
- [9] J. Ellis, J. L. Evans, S. Hor, N. Nagata, and K. A. Olive, *Planck-scale effects on nucleon decay in minimal supersymmetric $SU(5)$* , arXiv:2603.22418 [hep-ph] (2026).
- [10] J. Ehlers, F. A. E. Pirani, and A. Schild, *The geometry of free fall and light propagation*, in L. O’Raifeartaigh (ed.), *General Relativity: Papers in Honour of J. L. Synge*, Oxford University Press, 1972; republished in Gen. Rel. Grav. **44** (2012), 1587–1609.
- [11] J. T. Wheeler, *Geometry from geodesics: Fine-tuning Ehlers, Pirani, and Schild*, arXiv:2404.03815 [gr-qc] (2025).
- [12] N. Linnemann and J. Read, *Constructive axiomatics in spacetime physics Part I: Walkthrough to the Ehlers–Pirani–Schild axiomatisation*, Stud. Hist. Phil. Mod. Phys. **77** (2022), 1–17, arXiv:2112.14063 [gr-qc].
- [13] T. Jacobson, *Thermodynamics of spacetime: The Einstein equation of state*, Phys. Rev. Lett. **75** (1995), 1260–1263, arXiv:gr-qc/9504004.

- [14] M. Visser, *Sakharov's induced gravity: A modern perspective*, Mod. Phys. Lett. A **17** (2002), 977–992, arXiv:gr-qc/0204062.

- [15] C. Barceló, S. Liberati, and M. Visser, *Analogue gravity*, Living Rev. Relativ. **14**, 3 (2011), arXiv:gr-qc/0505065.
- [16] C. Rovelli and L. Smolin, *Spin networks and quantum gravity*, Phys. Rev. D **52** (1995), 5743–5759, arXiv:gr-qc/9505006.
- [17] L. Bombelli, J. Lee, D. Meyer, and R. D. Sorkin, *Space-time as a causal set*, Phys. Rev. Lett. **59** (1987), 521–524.
- [18] S. Surya, *The causal set approach to quantum gravity* Living Rev. Relativ. **22**, 5 (2019), arXiv:1903.11544 [gr-qc].
- [19] A. Perez, *The spin-foam approach to quantum gravity*, Living Rev. Relativ. **16**, 3 (2013), arXiv:1205.2019 [gr-qc].
- [20] T. Konopka, F. Markopoulou, and L. Smolin, *Quantum graphity*, arXiv:hep-th/0611197 (2006).
- [21] T. Padmanabhan, *Thermodynamical aspects of gravity: New insights*, Rep. Prog. Phys. **73** (2010), 046901, arXiv:0911.5004 [gr-qc].
- [22] B. S. DeWitt, *Quantum theory of gravity. I. The canonical theory* Phys. Rev. **160** (1967), 1113–1148.
- [23] M. Reuter, F.-E. Wolter, and N. Peinecke, *Laplace–Beltrami spectra as ‘Shape-DNA’ of surfaces and solids*, Comput.-Aided Des. **38** (2006), 342–366.
- [24] B. L. Hu, *Emergent/quantum gravity: Macro/micro structures of spacetime*, J. Phys.: Conf. Ser. **174** (2009), 012015, arXiv:0903.0878 [gr-qc].
- [25] I. Romero, *Variational principles for nonlinear Kirchhoff rods*, arXiv:1902.05726 [math-ph] (2019).
- [26] M. Gazzola, L. Heltai, A. G. S. Nair, and A. DeSimone, *Forward and inverse problems in the mechanics of soft filaments*, R. Soc. Open Sci. **5**:171628 (2018).
- [27] C. Alessi et al., *Rod models in continuum and soft robot control: A review*, arXiv:2407.05886 [cs.RO] (2024).

- [28] M. Bright, O. Anosova, and V. Kurlin, *A proof of the invariant-based formula for the linking number and its asymptotic behaviour*, arXiv:2011.04631 [math.GT] (2020).
- [29] R. Arnowitt, S. Deser, and C. W. Misner, *Republication of: The dynamics of general relativity*, Gen. Rel. Grav. **40** (2008), 1997–2027.
- [30] F. Mercati, *A shape dynamics tutorial*, arXiv:1409.0475 [gr-qc] (2014).

- [31] C. J. Isham, *Canonical quantum gravity and the problem of time*, in A. Ibort and M. A. Rodríguez (eds.), *Integrable Systems, Quantum Groups, and Quantum Field Theories*, NATO ASI Series C, vol. 409, Springer, Dordrecht, 1993, pp. 157–287, arXiv:gr-qc/9210011.

